ADVANCED COGNITIVE NEUROSCIENCE

MULTIVARIATE ANALYSIS, CONTINUED

LAURA BOCK PAULSEN
COGNITIVE SCIENCE, MSC
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HOUSEKEEPING

Is source reconstruction required for the report?

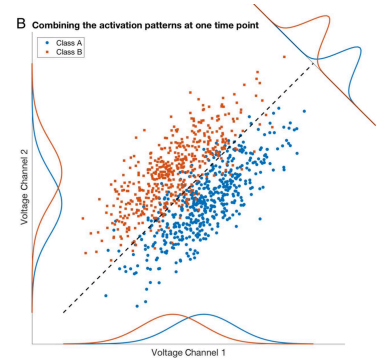
The report must include an introduction, based on relevant literature; operationalisable hypotheses; a clear methods section, which must describe the experiment and the data acquisition procedure; a succinct results section with relevant behavioural data, MEG data and source reconstructions; a discussion addressing limitations and the suitability of the method for investigating subjective experience. Data from all 8 participants must be used and at least one analysis must be based on multivariate statistics. A maximum of 6 pages per co-author. Code as appendix.

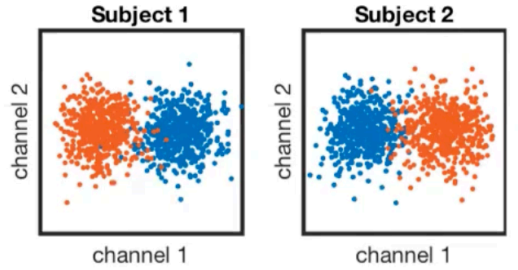
RECAP

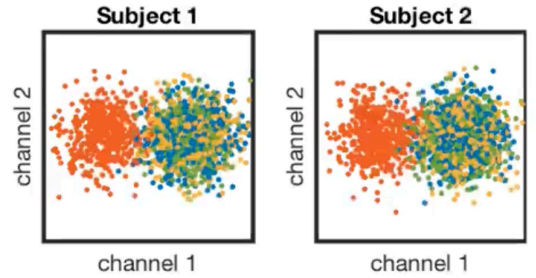
- Univariate vs **multivariate**
  - Univariate: No significant difference between **A** & **B** in either channel
  - Multiple variables: Information in the two channels combined about **A** & **B**

- Directional vs **non-directional**
  - Univariate: Subject 1 & channel 1 **A** < **B** + Subject 2 & channel 1 **A** > **B**
  - Multivariate analysis: For both subjects we would find that **A** ≠ **B**

- Activation vs **information**
  - Univariate: Increased brain activity for **B**, **C** & **D.**
    - Response reflects **B**, **C** & **D**
  - Multivariate: **A** is different from **B**, **C** & **D**
    - Response encodes information about **A**







DISCUSSION



Would you use multivariate or univariate analysis to answer each of these questions?

1. Does higher Gabor contrast increase activity in V1?
2. Is information about perceptual awareness represented in the occipital and frontal regions?
3. Is information about Gabor patch orientation represented in V1?
4. Does V1 activity differ between trials with high and low perceptual awareness?

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PHD STUDENT

WHICH VARIABLES COULD YOU DECODE IN OUR EXPERIMENT?

Classification

Regression



Kind of event (gabor, mask, button press)

Correct vs incorrect

PAS score

Orientation

Trial number

Contrast

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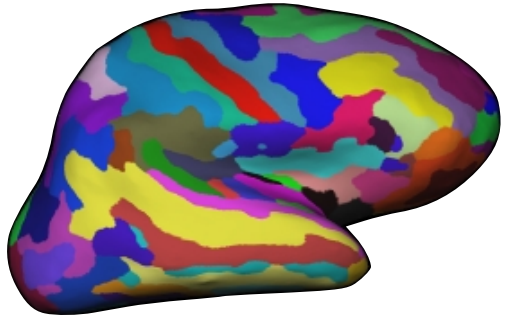
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WHAT SOURCE LABELS MAY BE INTERESTING?

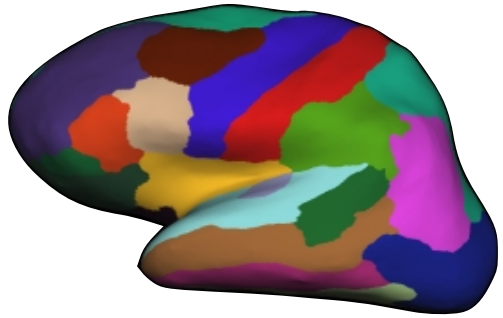


```
labels = mne.read_labels_from_annot(
    subject,
    subjects_dir=subjects_dir,
    parc='aparc.a2009s',  # or 'aparc'
    hemi="rh" # or “lh" or "both"
)
print(labels)
```

aparc.a2009s



aparc



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REGRESSING OUT THE EFFECT OF CONTRAST BEFORE DECODING

Contrast can potentially act as a confound in our decoding analysis

How can a confound affect decoding accuracy?

- Inflate decoding accuracy: the model exploits contrast as a proxy for the variable of interest (if contrast and PAS scores are correlated)
- Decrease decoding accuracy: variation in contrast adds noise/variability that obscures the information of interest.

We can (to some degree) regress out the effect of contrast on our MEG data.

Barnett et al (2024) *Identifying content-invariant neural signatures of perceptual vividness.*

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REGRESSING OUT EFFECTS OF OTHER VARIABLES

Basic idea: fit a linear model which predicts the MEG data from the contrast. Then we use the residuals from this model as input to our classifier!

Barnett et al (2024) *Identifying content-invariant neural signatures of perceptual vividness.*

REGRESSING OUT EFFECTS OF OTHER VARIABLES

Suppose we have *n* stimulus-locked epochs.

For each epoch, we extract the MEG signal from **all sensors** at **one time point** for all trials.

$$Y = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,M} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ y_{N,1} & y_{N,2} & \cdots & y_{N,M} \end{bmatrix}$$

Where $y_{N,M}$ is the amplitude in trial *N in* sensor *M*

REGRESSING OUT EFFECTS OF OTHER VARIABLES

For each trial we also know the contrast *C*.

$$\mathbf{C} = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{bmatrix}$$

We can then ask how much of the trial-to-trial variability in the MEG signal is explained by contrast

REGRESSING OUT EFFECTS OF OTHER VARIABLES

We then create our design matrix

$$X = \begin{bmatrix} 1 & c_1 \\ 1 & c_2 \\ \vdots & \vdots \\ 1 & c_N \end{bmatrix}$$

REGRESSING OUT EFFECTS OF OTHER VARIABLES

Using least squares, we estimate the coefficients to quantify the linear effect of the contrast on the signal at each sensor.

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$



What shape is β̂?

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REGRESSING OUT EFFECTS OF OTHER VARIABLES

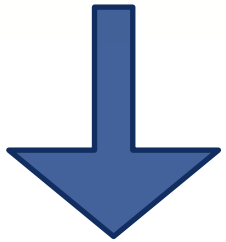
Next step is to predict the MEG component which is linearly explained by the contrast using the coefficients!

$$\hat{Y} = X\hat{\beta}$$

REGRESSING OUT EFFECTS OF OTHER VARIABLES

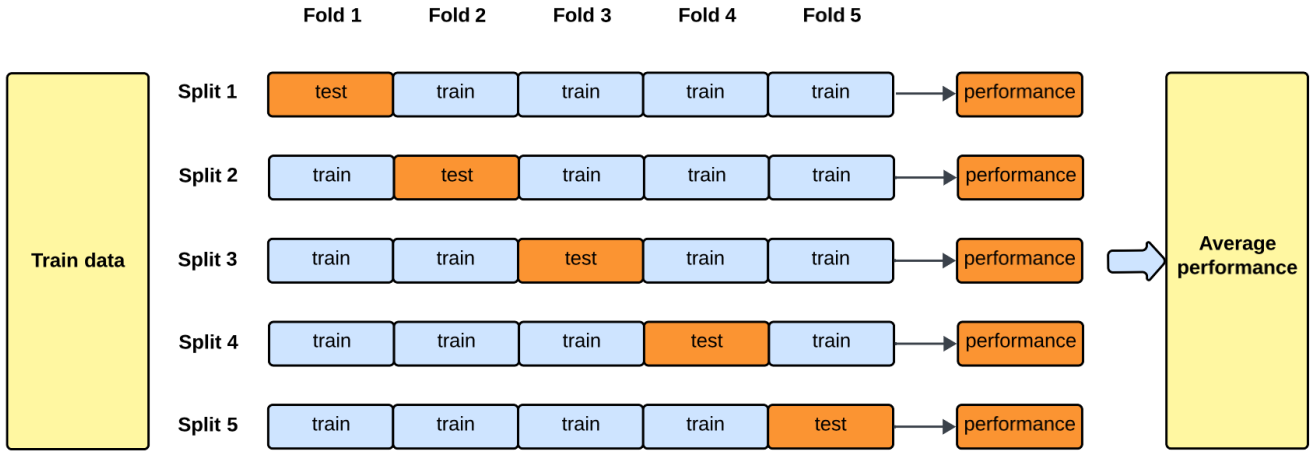
We can then remove this component from the original data to obtain the residual activity

$$Y_{\text{residual}} = Y - \hat{Y}$$



Input to our classifier

RECAP: CROSS-VALIDATION



https://towardsdatascience.com/how-to-cross-validation-with-time-series-data-9802a06272c6/

```
kfold = StratifiedKFold(5, shuffle=True)

pipeline = Pipeline(
    [("scaler", scaler), ("clf", clf)]
)


scores = cross_val_score(
    pipeline,
    X, y,
    cv=kfold
)

print(f"Mean accuracy = {np.mean(scores)}")
```

CROSS-VALIDATION AND REGRESSING OUT CONTRAST

Workflow

1. Fit regression on the training data
2. Regress contrast out of the training data
3. Apply the same coefficients to test data
4. Evaluate the model on the “residualised” test data

This is to avoid data leakage where information from the test set is used to estimate the regression coefficients

```
pipe = Pipeline([
   (“regress_out", RegressOutContrast(n_sensors=n_sensors)),
   ("scaler", scaler),
   ("clf", clf)
])
```